

❖ Title

GestuX Gloves: A Sign Language Vocalizer for Gesture-Based Communication Using Arduino Uno

❖ Introduction

GestuX Gloves is a wearable system designed to convert **sign language gestures** into **text and speech**, enabling effective communication for deaf and dumb individuals. The system focuses on gesture-based interaction, where hand movements are interpreted to convey meaningful messages. It aims to improve accessibility by allowing communication with people who are unfamiliar with sign language.

The idea is inspired by the project **SignAloud**, developed by **Thomas Pryor** and Navid Azodi at the **University of Washington**. Their work demonstrated that sign language can be translated into text and speech using wearable technology. However, such systems are complex and expensive, which limits their widespread adoption. This highlighted the need for a simpler and more practical vocalization approach.

The proposed system generates **predefined textual outputs** for common daily communication, which are presented through a **visual interface** on the glove. For audible communication, the same message can be accessed on a **connected mobile application**, where it is spoken in a selected language using the smartphone speaker. This allows the glove to function as a **vocalizer** without producing sound directly.

❖ Objectives

- To design a **wearable gesture-based system** for assisting communication using sign language.
- To display **predefined text messages** for common daily interactions through a visual interface.
- To support **speech output** using a connected smartphone application.
- To provide a **gesture-based vocalization approach** suitable for real-world communication.
- To enable communication using **multiple predefined phrases** for daily use.
- To develop a system that is **simple, portable, and easy to use**.

❖ Scope

- The project focuses on **gesture-based vocalization** for basic daily communication.
- It supports a set of **predefined phrases** to ensure reliable message interpretation.
- The system is intended for use in **educational, medical, and social environments**.
- The design emphasizes **portability and ease of interaction**.

- The project allows **future enhancement**, such as adding more gestures or language options.

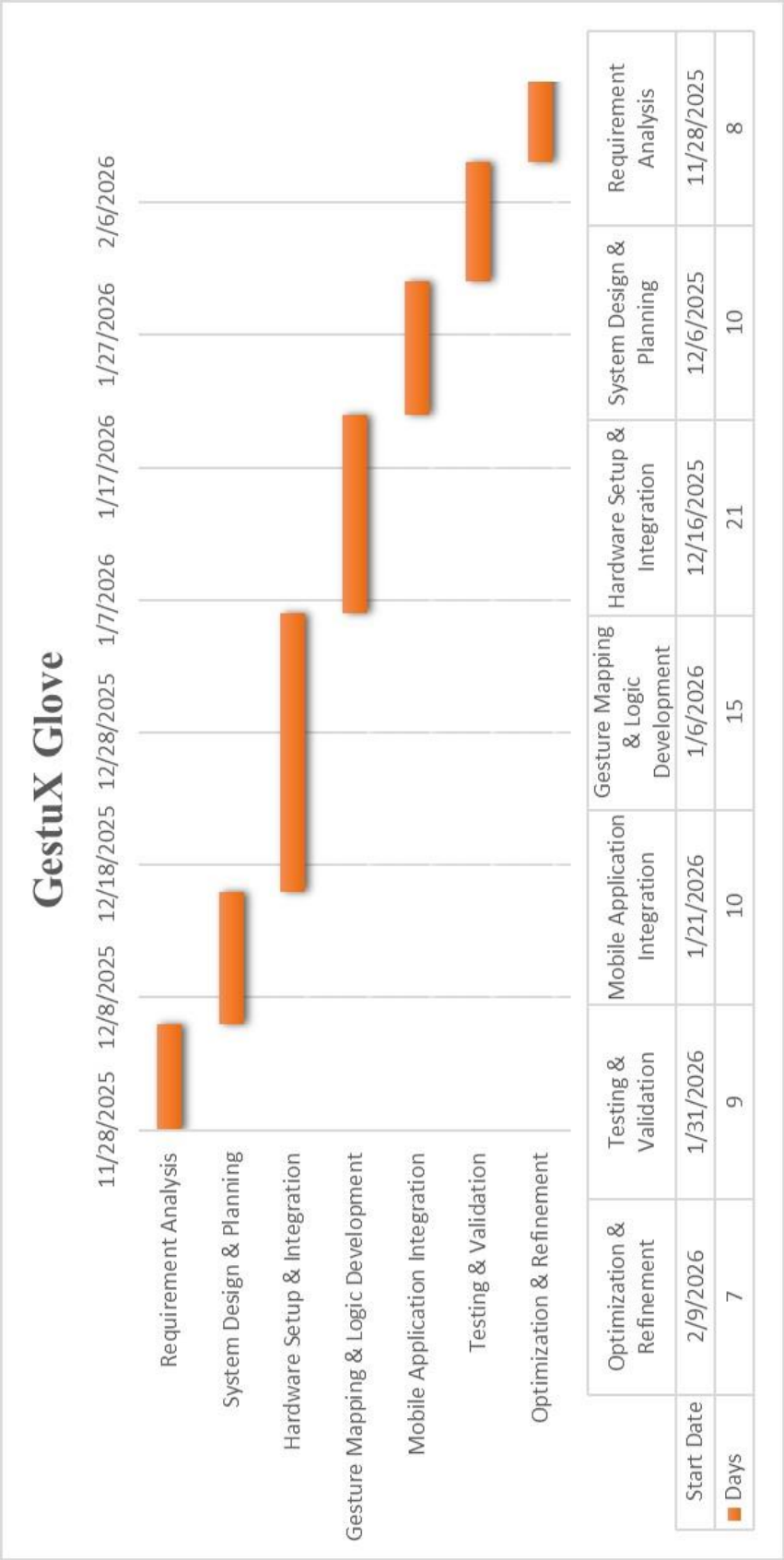
❖ **Methodology**

- **Requirement Analysis:** Identifying communication needs of deaf and dumb users and reviewing existing gesture-based systems.
- **System Planning:** Designing a wearable gesture-based vocalizer using Arduino Uno with a focus on simplicity.
- **Gesture Interpretation:** Mapping recognized hand gestures to predefined text messages.
- **Visual Output Generation:** Displaying the identified message through a visual interface on the glove.
- **Mobile Integration:** Transferring message data wirelessly to a smartphone application.
- **Speech Output:** Converting the received text into audible speech using the mobile device.

❖ **Tools and Technologies**

- ❖ **Arduino Uno:** Used as the main controller to process gesture inputs and manage display and communication outputs.
- ❖ **MPU6050 (Accelerometer/Gyroscope):** Used to detect hand orientation and tilt for motion-based gesture recognition.
- ❖ **HC-05 Bluetooth Module:** Used to enable wireless data transfer between the glove and the smartphone.
- ❖ **16×2 LCD Display:** Used to show predefined text messages for visual confirmation.
- ❖ **Basic Contact-Based Input Logic:** Used to detect finger positions in a simple and practical manner.
- ❖ **MIT App Inventor:** Used to develop a mobile application that converts received text into speech using the **Android Text-to-Speech (TTS) engine**.
- ❖ **Arduino IDE:** Used for writing, compiling, and uploading the program to the Arduino board.

❖ Timeline (Gantt Chart Format)



Phase	Start Date	End Date	Days
Requirement Analysis	2025-11-28	2025-12-05	8
System Design & Planning	2025-12-06	2025-12-15	10
Hardware Setup & Integration	2025-12-16	2026-01-05	21
Gesture Mapping & Logic Development	2026-01-06	2026-01-20	15
Mobile Application Integration	2026-01-21	2026-01-30	10
Testing & Validation	2026-01-31	2026-02-08	9
Optimization & Refinement	2026-02-09	2026-02-15	7

❖ Resources

- **Hardware Components:** Arduino Uno, sensors, display module, Bluetooth module, power supply, and basic electronic accessories.
- **Software Tools:** Arduino IDE for programming and MIT App Inventor for mobile application development.
- **Development Platform:** Android smartphone for application testing and speech output.
- **Reference Material:** Research papers, technical documentation, and online resources related to sign language and gesture recognition.
- **Human Resources:** Guidance from project mentor and feedback from peers during development and testing.

❖ Expected Outcomes

- A **functional gesture-based vocalizer** capable of converting sign language gestures into readable text.
- Clear **visual display of predefined messages** on the glove for immediate understanding.
- **Audible speech output** generated through a connected smartphone with language selection support.
- A **portable wearable system** suitable for everyday gesture-based communication.
- Improved **communication accessibility** for deaf and dumb individuals in social and educational settings.
- A **scalable framework** that allows future expansion with additional gestures or phrases.

❖ References

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